

Mid-term examination on Logic Circuit Design (Group 2)

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1 Balanced Ternary

Question. Write a C function which converts an integer of type `int` into its balanced ternary representation as an array of `int`. The prototype is

```
int* from10to3b (int dec);
```

The result of the function is an array of integers where the first element is the size of the balanced ternary representation and the remaining elements are the digits (also known as *trits*), the least significant being the last. For example, `from10to3b(17)` returns an array

4	1	-1	0	-1
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because $17_{10} = 1\bar{1}0\bar{1}_{3b}$.

Hint. First, compute the length of the balanced ternary representation by using the same algorithm used for the conversion. Then allocate an array and finally run the conversion algorithm as usual. Try some examples, like -17 . In C, the sign of `x % y` is the sign of `x`. You need to include `stdio.h` and `stdlib.h` (for `malloc`).

Question. If an `int` is stored as a 2-complement 32-bit machine word, what are the smallest and largest numbers your program can convert?